EXPERIMENT 05

Exploratory Data Analysis – Data Inspection, Conditional Subsetting, Descriptive Statistics, and Visualization using Python

1. AIM

To perform Exploratory Data Analysis (EDA) by inspecting DataFrames, filtering data using single and multiple conditions, calculating measures of central tendency and dispersion, and visualizing data distributions using Pandas, Seaborn, and Matplotlib.

2. SOFTWARE & LIBRARIES REQUIRED

Tool / Library	Purpose / Description
Python 3.x	Programming Language
Pandas	Data manipulation and analysis library
Matplotlib	Data visualization library
Seaborn	Statistical data visualization library based on Matplotlib
Jupyter Notebook / VS Code	Development Environment

3. PROCEDURE

Step 1: Import Libraries & Load Dataset – Import required libraries (pandas, matplotlib.pyplot, seaborn) and read the CSV dataset into a DataFrame.

Step 2: Inspect DataFrame – Perform structural inspection using head(), tail(), shape, columns.tolist(), dtypes, info(), and describe().

Step 3: Subset & Filter Data – Extract specific columns (gender, math score) and apply single/multiple conditional filters (e.g., math score > 70, female students with master's degree).

Step 4: Calculate Descriptive Statistics – Compute measures of central tendency (mean, median, mode) and dispersion (range, variance, standard deviation).

Step 5: Data Visualization – Generate visualizations including bar charts, histograms, boxplots, and histograms overlaid with central tendency lines.

4. PYTHON CODE IMPLEMENTATION

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns

# Load dataset
df = pd.read_csv('StudentsPerformance.csv')

# ------------------------------------------------------------
# 1. VIEWING AND INSPECTING DATAFRAMES
# ------------------------------------------------------------
print("=== FIRST 5 ROWS ===")
print(df.head())

print("\n=== LAST 5 ROWS ===")
print(df.tail())

print("\n=== SHAPE OF DATAFRAME ===")
print(df.shape)

print("\n=== COLUMN NAMES ===")
print(df.columns.tolist())

print("\n=== DATA TYPES ===")
print(df.dtypes)

print("\n=== DATAFRAME SUMMARY ===")
df.info()

print("\n=== SUMMARY STATISTICS ===")
print(df.describe())

# ------------------------------------------------------------
# 2. FILTERING AND SUBSETTING DATA
# ------------------------------------------------------------
# Select only 'gender' and 'math score' columns
gender_math = df[['gender', 'math score']]
print("\n=== GENDER AND MATH SCORE ===")
print(gender_math.head())

# Filter: Math score > 70
math_above_70 = df[df['math score'] > 70]
print("\n=== MATH SCORE > 70 ===")
print(math_above_70.head())

# Filter: Female students with math score > 70
female_math_above_70 = df[(df['gender'] == 'female') & (df['math score'] > 70)]
print("\n=== FEMALE STUDENTS WITH MATH > 70 ===")
print(female_math_above_70.head())

# ------------------------------------------------------------
# 3. DESCRIPTIVE STATISTICS
# ------------------------------------------------------------
# Central Tendency
math_mean = df['math score'].mean()
math_median = df['math score'].median()
```

```
math_mode = df['math score'].mode()[0]

# Dispersion
math_min = df['math score'].min()
math_max = df['math score'].max()
math_range = math_max - math_min
math_var = df['math score'].var()
math_std = df['math score'].std()

print("\n=== DESCRIPTIVE STATISTICS FOR MATH SCORE ===")
print(f"Mean: {math_mean:.2f}")
print(f"Median: {math_median}")
print(f"Mode: {math_mode}")
print(f"Range: {math_range} (Min: {math_min}, Max: {math_max})")
print(f"Variance: {math_var:.2f}")
print(f"Standard Deviation: {math_std:.2f}")

# ------------------------------------------------------------
# 4. DATA VISUALIZATION
# ------------------------------------------------------------
plt.style.use('seaborn-v0_8-whitegrid')

# Task: Visualize only female students with master's degree parents
female_masters = df[(df['gender'] == 'female') & (df['parental level of education'] ==
"master's degree")]
plt.figure(figsize=(6, 4))
sns.barplot(data=female_masters, x='gender', y='math score', palette='Set2')
plt.title("Math Scores of Female Students with Master's Degree Parents")
plt.show()

# 1. Bar Chart: Average scores per subject
subject_means = df[['math score', 'reading score', 'writing score']].mean()
plt.figure(figsize=(8, 5))
subject_means.plot(kind='bar', color=['#4C72B0', '#DD8452', '#55A868'])
plt.title("Average Scores per Subject")
plt.ylabel("Average Score")
plt.xticks(rotation=0)
plt.show()

# 2. Histogram: Distribution of math scores
plt.figure(figsize=(8, 5))
sns.histplot(df['math score'], kde=True, color='skyblue')
plt.title("Distribution of Math Scores")
plt.xlabel("Math Score")
plt.ylabel("Frequency")
plt.show()

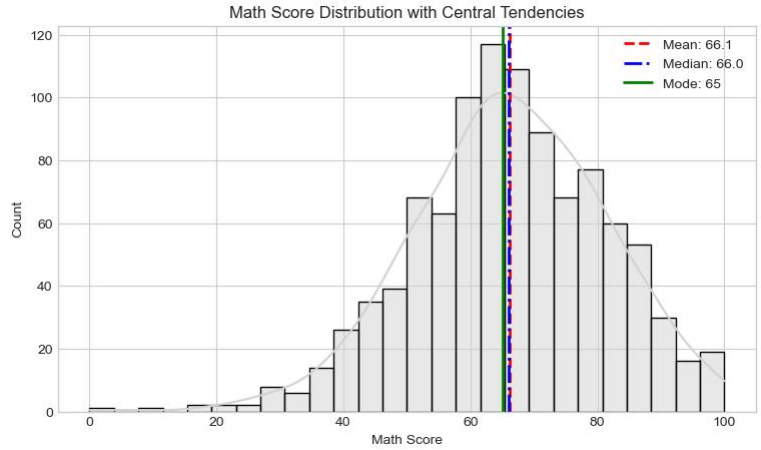
# 3. Boxplot: Spread of math scores
plt.figure(figsize=(6, 4))
sns.boxplot(y=df['math score'], color='lightgreen')
plt.title("Spread of Math Scores (Boxplot)")
plt.ylabel("Math Score")
plt.show()
```

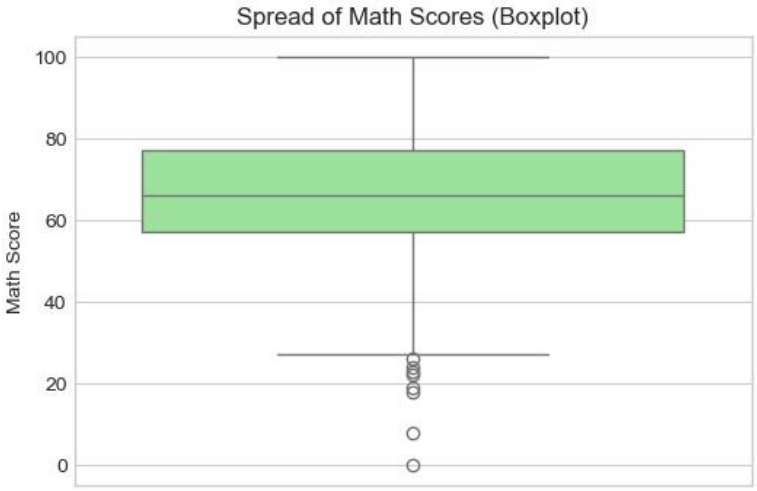
```
# 4. Plot Histogram with Mean, Median, and Mode Lines
plt.figure(figsize=(9, 5))
sns.histplot(df['math score'], kde=True, color='lightgray')
plt.axvline(math_mean, color='red', linestyle='--', linewidth=2, label=f'Mean:
{math_mean:.1f}')
plt.axvline(math_median, color='blue', linestyle='-.', linewidth=2, label=f'Median:
{math_median}')
plt.axvline(math_mode, color='green', linestyle='-', linewidth=2, label=f'Mode:
{math_mode}')
plt.title("Math Score Distribution with Central Tendencies")
plt.xlabel("Math Score")
plt.ylabel("Count")
plt.legend()
plt.show()
```

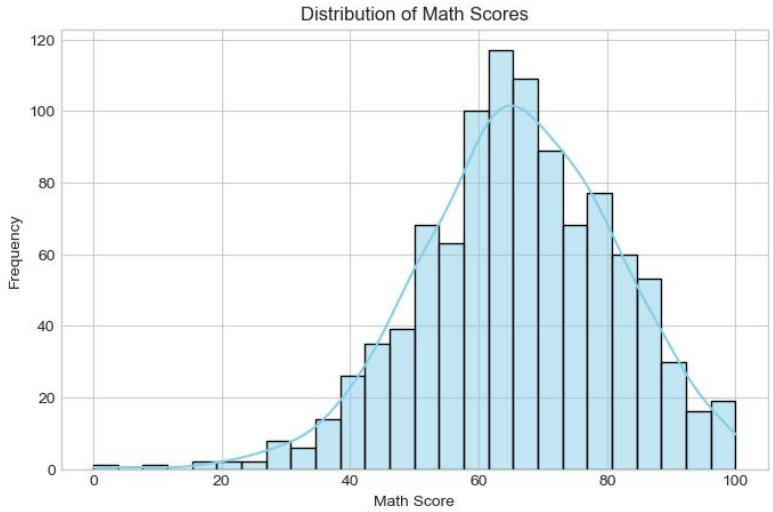
5. SUMMARY OF PANDAS & SEABORN METHODS USED

Method / Syntax	Description / Function
df.head() / df.tail()	Displays first / last 5 rows of the DataFrame
df.shape	Returns tuple representing dimensions (rows, columns)
df.columns.tolist()	Converts DataFrame column headers into a standard list
df.dtypes / df.info()	Inspects data types and checks missing values/memory usage
df.describe()	Generates overall summary statistics for numerical columns
df[(cond1) & (cond2)]	Subsets DataFrame using logical AND conditional operations
.mean(), .median(), .mode()	Calculates measures of central tendency
.var(), .std(), max() - min()	Calculates measures of dispersion
sns.histplot() / sns.boxplot()	Visualizes data distribution and spread using Seaborn

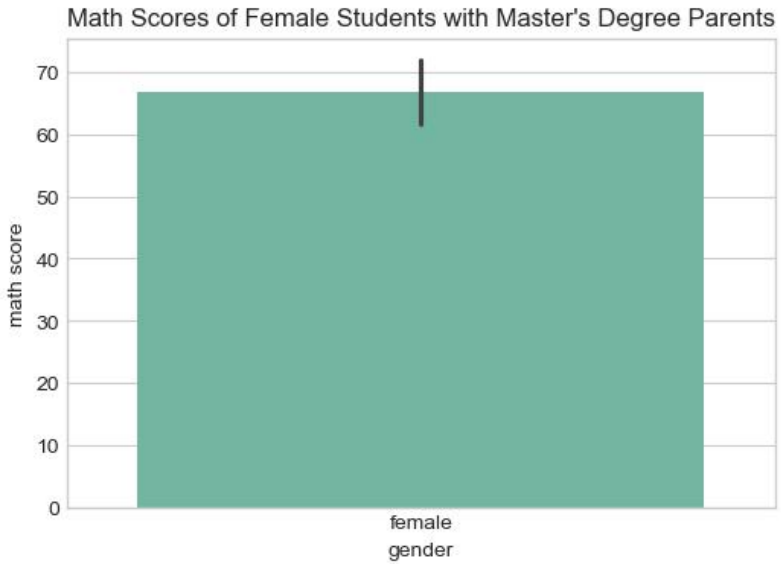
6. OUTPUT


Math Score Distribution with Central Tendencies
Mean: 66.1
Median: 66.0
Mode: 65
Count
Math Score


Spread of Math Scores (Boxplot)
Math Score


Distribution of Math Scores
Frequency
Math Score


Average Scores per Subject
Average Score
0
10
20
30
40
50
60
70
math score
reading score
writing score


Math Scores of Female Students with Master's Degree Parents
math score
0
10
20
30
40
50
60
70
female
gender

```
=== FIRST 5 ROWS ===
   gender race/ethnicity parental level of education         lunch  \
0  female        group B           bachelor's degree      standard
1  female        group C                some college      standard
2  female        group B             master's degree      standard
3    male        group A          associate's degree  free/reduced
4    male        group C                some college      standard

  test preparation course  math score  reading score  writing score
0                    none          72             72             74
1               completed          69             90             88
2                    none          90             95             93
3                    none          47             57             44
4                    none          76             78             75

=== LAST 5 ROWS ===
      gender race/ethnicity parental level of education     lunch  \
1000    male        group D                some college  standard
1001    male        group C           associate's degree  standard
1002  female        group B            bachelor's degree  standard
1003    male        group E             some high school  standard
1004    male        group C            bachelor's degree  standard

     test preparation course  math score  reading score  writing score
1000                    none          76             64             66
1001                    none          46             43             42
1002                    none          67             86             83
1003                    none          92             87             78
1004               completed          83             82             84

=== SHAPE OF DATAFRAME ===
(1005, 8)
```

```
=== DATA TYPES ===
gender                         object
race/ethnicity                 object
parental level of education    object
lunch                          object
test preparation course        object
math score                      int64
reading score                   int64
writing score                   int64
dtype: object

=== DATAFRAME SUMMARY ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1005 entries, 0 to 1004
Data columns (total 8 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   gender                       1005 non-null   object
 1   race/ethnicity               1005 non-null   object
 2   parental level of education  998 non-null    object
 3   lunch                        1005 non-null   object
 4   test preparation course      1005 non-null   object
 5   math score                   1005 non-null   int64
 6   reading score                1005 non-null   int64
 7   writing score                1005 non-null   int64
dtypes: int64(3), object(5)
memory usage: 62.9+ KB

=== SUMMARY STATISTICS ===
        math score  reading score  writing score
count  1005.000000    1005.000000    1005.000000
mean     66.122388      69.185075      68.066667
std      15.173234      14.614215      15.199095
min       0.000000      17.000000      10.000000
25%      57.000000      59.000000      58.000000
50%      66.000000      70.000000      69.000000
75%      77.000000      80.000000      79.000000
max     100.000000     100.000000     100.000000

=== GENDER AND MATH SCORE ===
   gender  math score
0  female          72
1  female          69
2  female          90
3    male          47
4    male          76
```

```
=== MATH SCORE > 70 ===
   gender race/ethnicity parental level of education     lunch  \
0  female        group B           bachelor's degree  standard
2  female        group B             master's degree  standard
4    male        group C                some college  standard
5  female        group B          associate's degree  standard
6  female        group B                some college  standard

  test preparation course  math score  reading score  writing score
0                    none          72             72             74
2                    none          90             95             93
4                    none          76             78             75
5                    none          71             83             78
6               completed          88             95             92

=== FEMALE STUDENTS WITH MATH > 70 ===
    gender race/ethnicity parental level of education     lunch  \
0   female        group B           bachelor's degree  standard
2   female        group B             master's degree  standard
5   female        group B          associate's degree  standard
6   female        group B                some college  standard
36  female        group D          associate's degree  standard

   test preparation course  math score  reading score  writing score
0                    none          72             72             74
2                    none          90             95             93
5                    none          71             83             78
6               completed          88             95             92
36                   none          74             81             83

=== DESCRIPTIVE STATISTICS FOR MATH SCORE ===
Mean: 66.12
Median: 66.0
Mode: 65
Range: 100 (Min: 0, Max: 100)
Variance: 230.23
Standard Deviation: 15.17
```

7. **RESULT**

- Performed EDA by inspecting DataFrame structure, filtering conditional subsets, and calculating central tendency and dispersion.
- Visualized dataset distributions using bar charts, boxplots, and histograms marked with mean, median, and mode lines.